6

Missing Data

Don't play what's there; play what's not there.

—Miles Davis

In Part II, we showed that, if we observe a sufficient number of independent and identically distributed (i.i.d.) draws of a random vector, we can estimate any feature of its distribution to arbitrary precision. Given any finite number of such observations, we can both estimate these features and, furthermore, estimate our degree of uncertainty about these estimates. Thus, if we could always observe i.i.d. draws of any random vector whose distribution we might be interested in, this book could have concluded with Chapter 5.

Unfortunately, we often want to learn about the distributions of random vectors that are *not* directly observable. What can we learn about the distribution of an *unobservable* random vector, given full knowledge of the distribution of a related *observable* random vector? This question is the core of nonparametric statistical *identification*, an idea that we will explore in Part III. The answer will invariably depend on what assumptions we are willing to make about the relationship between the observable and unobservable distributions. This notion can be formalized in the following definition of identification:

Definition 6.0.1. *Point Identification*
Suppose we have complete knowledge of the joint CDF, F, of some observable random vector. Then a statistical functional of some unobservable random vector with joint CDF F_u, $T(F_u)$, is *point identified* by a set of assumptions A_u if there exists one and only one value of $T(F_u)$ that is logically compatible with F given A_u.[1]

1 The notation F_u and A_u will not be used again.

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In this chapter, we illustrate the basic concepts of identification by considering a simple but common problem: missing data. Readers will notice that sections in this chapter bear a remarkable resemblance to those in Chapter 7, *Causal Inference*. This is not a mistake. Rather, we will show that causal inference can viewed as a missing data problem (see further discussion in Section 7.1.2). Consequently, the tools we use for identification and estimation with missing data are the same tools we will use with missing potential outcomes. To highlight these parallels, section numbers in brackets will reference the analogous section in Chapter 7. However, given the central role that causal claims play in much of empirical social scientific inquiry, Chapter 7 will go into greater detail than the relatively abbreviated, but foundational, treatment of missing data in this chapter.

6.1 IDENTIFICATION WITH MISSING DATA [7.1]

What can we learn about the distribution of a random variable of interest from a random sample when some values in the sample are missing? For example, if we ran a survey asking people whom they voted for, some people might not respond to the vote choice question. We want to know the population distribution of outcomes for this question, but we do not observe the outcomes for some people.

Suppose that we are interested in estimating the expected value (or the full distribution) of some random variable Y_i.[2] However, we do not observe Y_i for any unit in our sample that does not respond. Formally, let R_i be an indicator for whether or not the outcome Y_i for unit i is observed (that is, $R_i = 1$ if unit i responded and $R_i = 0$ otherwise), and let Y_i^* denote the censored version of Y_i. Then we can define the following model of missing data:

Definition 6.1.1. *Stable Outcomes Model*
For an outcome variable Y_i, the *stable outcomes model* is

$$Y_i^* = \begin{cases} -99 & : \quad R_i = 0 \\ Y_i & : \quad R_i = 1 \end{cases}$$
$$= Y_i R_i + (-99)(1 - R_i),$$

2 Since we are assuming i.i.d. sampling, our inferences do not depend on the unit subscript i. That is, learning about the random variable Y is equivalent to learning about the population distribution of Y_i values. However, whereas in Part II we largely treated i.i.d. random variables simply as abstract mathematical objects, in Part III i.i.d. random variables represent individual units in a random sample from some actual population. We therefore use the subscript i to emphasize that we are modeling the (observed and unobserved) features of sampled units from a population as i.i.d. random variables.

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where Y_i^* is the censored version of Y_i and R_i is an indicator for whether Y_i is observed for unit i.[3]

We refer to this as the *stable outcomes* framework because the underlying Y_i for unit i is stable in that it does not depend on how the question was asked, who was asked, or who responded.

Suppose that our inferential target is the expected value of Y_i, $\mathrm{E}[Y_i]$. We do *not* directly observe i.i.d. draws of Y_i; we only observe i.i.d. draws of the random vector (Y_i^*, R_i). Thus, without making further assumptions about the joint distribution of Y_i and R_i, we cannot directly estimate $\mathrm{E}[Y_i]$. So the question is: given full knowledge of the joint distribution of (Y_i^*, R_i), what assumptions yield what information about $\mathrm{E}[Y_i]$?

6.1.1 Bounds [7.1.3]

Let us start with a minimal assumption. Suppose that Y_i is *bounded*: $\mathrm{Supp}[Y_i] \subseteq [a, b]$, for some $a \leq b$. Under this assumption, we can derive *sharp bounds*—that is, bounds that cannot be improved upon without further assumptions—on $\mathrm{E}[Y_i]$.[4]

We can compute sharp bounds by imputing the "worst-case scenario" for all missing data. To compute a lower bound, assume that all missing values of Y_i equal the lowest possible value, a. For an upper bound, assume that all missing values of Y_i equal the highest possible value, b.

Example 6.1.2. *Bounding the Expected Value*
Suppose that Y_i is binary: $\mathrm{Supp}[Y_i] \subseteq \{0, 1\} \subseteq [0, 1]$. We might observe the following data:

Unit	Y_i^*	R_i
1	1	1
2	−99	0
3	1	1
4	0	1
5	1	1
6	−99	0

3 The value −99 is commonly used in datasets to denote missing data, but the specific number used is irrelevant.

4 These bounds are often referred to as *Manski bounds* due to his seminal work on the topic. See, e.g., Manski (2003).

Given the assumption of stable outcomes, this implies

Unit	Y_i	R_i
1	1	1
2	?	0
3	1	1
4	0	1
5	1	1
6	?	0

For units 2 and 6, we only know that $Y_i \in \{0,1\}$. To estimate a lower bound, we replace these missing values with the lowest possible value:

Unit	Y_i	R_i
1	1	1
2	0	0
3	1	1
4	0	1
5	1	1
6	0	0

Then, using the sample mean as a plug-in estimator for the expected value, our estimated lower bound for $\mathrm{E}[Y_i]$ is $\frac{1}{2}$. Likewise, to estimate an upper bound, we replace these missing values with the lowest possible values:

Unit	Y_i	R_i
1	1	1
2	1	0
3	1	1
4	0	1
5	1	1
6	1	0

Thus, our estimated upper bound for $\mathrm{E}[Y_i]$ is $\frac{5}{6}$. △

More generally, we can write down exact sharp bounds for $\mathrm{E}[Y_i]$ in terms of the joint distribution of (Y_i^*, R_i), which can, in theory, be estimated to arbitrary precision.

Theorem 6.1.3. *Sharp Bounds for the Expected Value*
Let Y_i and R_i be random variables with $\mathrm{Supp}[Y_i] \subseteq [a,b]$ and $\mathrm{Supp}[R_i] = \{0,1\}$, and let $Y_i^* = Y_i R_i + (-99)(1-R_i)$. Then

$$\begin{aligned}\mathrm{E}[Y_i] \in \Big[&\mathrm{E}[Y_i^* \mid R_i = 1]\Pr[R_i = 1] + a\Pr[R_i = 0],\\ &\mathrm{E}[Y_i^* \mid R_i = 1]\Pr[R_i = 1] + b\Pr[R_i = 0]\Big].\end{aligned}$$

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Proof: By the Law of Iterated Expectations,

$$\mathrm{E}[Y_i] = \mathrm{E}[Y_i \,|\, R_i = 1]\Pr[R_i = 1] + \mathrm{E}[Y_i \,|\, R_i = 0]\Pr[R_i = 0].$$

Then, since $Y_i^* = Y_i R_i + (-99)(1 - R_i)$,

$$\begin{aligned}
\mathrm{E}[Y_i^* \,|\, R_i = 1] &= \mathrm{E}[Y_i R_i + (-99)(1 - R_i) \,|\, R_i = 1] \\
&= \mathrm{E}[Y_i \cdot 1 + (-99) \cdot 0 \,|\, R_i = 1] \\
&= \mathrm{E}[Y_i \,|\, R_i = 1].
\end{aligned}$$

Thus,

$$\mathrm{E}[Y_i] = \mathrm{E}[Y_i^* \,|\, R_i = 1]\Pr[R_i = 1] + \mathrm{E}[Y_i \,|\, R_i = 0]\Pr[R_i = 0].$$

Since $\mathrm{Supp}[Y_i] \subseteq [a, b]$, $\mathrm{E}[Y_i \,|\, R_i = 0] \geq a$, and therefore

$$\begin{aligned}
\mathrm{E}[Y_i] &= \mathrm{E}[Y_i^* \,|\, R_i = 1]\Pr[R_i = 1] + \mathrm{E}[Y_i \,|\, R_i = 0]\Pr[R_i = 0] \\
&\geq \mathrm{E}[Y_i^* \,|\, R_i = 1]\Pr[R_i = 1] + a\Pr[R_i = 0].
\end{aligned}$$

Likewise, since $\mathrm{Supp}[Y_i] \subseteq [a, b]$, $\mathrm{E}[Y_i \,|\, R_i = 0] \leq b$, and therefore

$$\begin{aligned}
\mathrm{E}[Y_i] &= \mathrm{E}[Y_i^* \,|\, R_i = 1]\Pr[R_i = 1] + \mathrm{E}[Y_i \,|\, R_i = 0]\Pr[R_i = 0] \\
&\leq \mathrm{E}[Y_i^* \,|\, R_i = 1]\Pr[R_i = 1] + b\Pr[R_i = 0].
\end{aligned}$$

Thus,

$$\begin{aligned}
\mathrm{E}[Y_i] \in \big[&\mathrm{E}[Y_i^* \,|\, R_i = 1]\Pr[R_i = 1] + a\Pr[R_i = 0], \\
&\mathrm{E}[Y_i^* \,|\, R_i = 1]\Pr[R_i = 1] + b\Pr[R_i = 0]\big]. \quad \square
\end{aligned}$$

One feature of these bounds is that their width is proportional to the amount of missing data; fundamental uncertainty grows as we observe less of the distribution of Y_i.

Estimation of these bounds follows from the plug-in principle (Section 3.3): substituting sample means for expected values. The following theorem establishes that the plug-in estimator of the bounds is unbiased and consistent.

Theorem 6.1.4. *Estimating Sharp Bounds for the Expected Value*
Let Y_i and R_i be random variables with $\mathrm{Supp}[Y_i] \subseteq [a, b]$ and $\mathrm{Supp}[R_i] = \{0, 1\}$, and let $Y_i^* = Y_i R_i + (-99)(1 - R_i)$. Then, given n i.i.d. observations of (Y_i^*, R_i), the plug-in estimator

$$\frac{1}{n}\sum_{i=1}^{n}\big(Y_i^* R_i + a(1 - R_i)\big)$$

is unbiased and consistent for the lower bound for $\mathrm{E}[Y_i]$. Likewise, the plug-in estimator

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$$\frac{1}{n}\sum_{i=1}^{n}\big(Y_i^* R_i + b(1-R_i)\big)$$

is unbiased and consistent for the upper bound for $\mathrm{E}[Y_i]$.

Proof: Let $Y_i^L = Y_i^* R_i + a(1-R_i)$, so that

$$\overline{Y_i^L} = \frac{1}{n}\sum_{i=1}^{n}\big(Y_i^* R_i + a(1-R_i)\big).$$

By Theorem 3.2.3 (*The Expected Value of the Sample Mean is the Population Mean*) and the Weak Law of Large Numbers (WLLN), $\overline{Y_i^L}$ is unbiased and consistent for $\mathrm{E}[Y_i^L]$, and by the Law of Iterated Expectations,

$$\begin{aligned}
\mathrm{E}\big[Y_i^L\big] &= \mathrm{E}\big[Y_i^L \,\big|\, R_i = 1\big]\Pr[R_i = 1] + \mathrm{E}\big[Y_i^L \,\big|\, R_i = 0\big]\Pr[R_i = 0] \\
&= \mathrm{E}\big[Y_i^* R_i + a(1-R_i) \,\big|\, R_i = 1\big]\Pr[R_i = 1] \\
&\quad + \mathrm{E}\big[Y_i^* R_i + a(1-R_i) \,\big|\, R_i = 0\big]\Pr[R_i = 0] \\
&= \mathrm{E}\big[Y_i^* \cdot 1 + a \cdot 0 | R_i = 1\big]\Pr\big[R_i = 1\big] \\
&\quad + \mathrm{E}\big[Y_i^* \cdot 0 + a \cdot 1 | R_i = 0\big]\Pr\big[R_i = 0\big] \\
&= \mathrm{E}\big[Y_i^* | R_i = 1\big]\Pr\big[R_i = 1\big] + a\Pr\big[R_i = 0\big].
\end{aligned}$$

Thus, $\overline{Y_i^L}$ is unbiased and consistent for $\mathrm{E}[Y_i^* | R_i = 1]\Pr[R_i = 1] + a\Pr[R_i = 0]$.

Similarly, let $Y_i^U = Y_i^* R_i + b(1-R_i)$, so that

$$\overline{Y_i^U} = \frac{1}{n}\sum_{i=1}^{n}\big(Y_i^* R_i + b(1-R_i)\big).$$

Then, by the same logic, it follows that $\overline{Y_i^U}$ is unbiased and consistent for $\mathrm{E}[Y_i^* | R_i = 1]\Pr[R_i = 1] + b\Pr[R_i = 0]$. □

Note that these plug-in estimators will reproduce our bounds estimate of $[\frac{1}{2}, \frac{5}{6}]$ from Example 6.1.2 (*Missing Data with Binary Outcomes*). Under the usual plug-in regularity conditions, asymptotically valid confidence intervals for each bound can be computed using the bootstrap.

Without further assumptions, these bounds are the absolute best we can do. Thus, under these minimal assumptions, $\mathrm{E}[Y_i]$ is *partially identified* (or *set identified*) as opposed to *point identified*. That is, even with full knowledge of the joint distribution of (Y_i^*, R_i), we can do no better than the above bounds. We are left with an interval of plausible values, none of which can be logically ruled out given the assumptions that we have imposed. We cannot identify

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the exact value of $\mathrm{E}[Y_i]$. Adding more assumptions inherently reduces the credibility of our estimate. With stronger assumptions, we might be able to narrow the interval, perhaps even to a point. But the cost is that our estimate becomes less believable. Manski (2003) refers to this as the "Law of Decreasing Credibility."

Sometimes, we should not feel comfortable strengthening our assumptions. Bounded support of outcomes may be the only assumption that we are willing to impose given what we know—or, more to the point, what we do not know—about how our data were generated. If the bounds are not good enough, we need further information about the generative process—in this case, the determinants of non-response—to justify the stronger (that is, more restrictive) assumptions required to obtain a more precise estimate.

6.1.2 Missing Completely at Random [7.1.4]

We can consider one of these stronger assumptions. Suppose that the data are *missing completely at random*. This is generally considered to be one of the strongest of all the nonparametric assumptions that we could impose.

Definition 6.1.5. *Missing Completely at Random (MCAR)*
Let Y_i and R_i be random variables with $\mathrm{Supp}[R_i] = \{0, 1\}$, and let $Y_i^* = Y_i R_i + (-99)(1 - R_i)$. Then Y_i is *missing completely at random* if the following conditions hold:

- $Y_i \perp\!\!\!\perp R_i$ (independence of outcome and response).
- $\Pr[R_i = 1] > 0$ (nonzero probability of response).

In other words, the distribution of outcomes for the population of people who respond is the same as the distribution of outcomes for the population of people who do not respond. The MCAR assumption is sufficient to point identify $\mathrm{E}[Y_i]$, as we can then write down an exact, unique expression for $\mathrm{E}[Y_i]$ in terms of the joint distribution of observables, (Y_i^*, R_i).

Theorem 6.1.6. *Expected Value under MCAR*
Let Y_i and R_i be random variables with $\mathrm{Supp}[R_i] = \{0, 1\}$, and let $Y_i^* = Y_i R_i + (-99)(1 - R_i)$. Then, if Y_i is missing completely at random,

$$\mathrm{E}[Y_i] = \mathrm{E}[Y_i^* | R_i = 1].$$

Proof: By independence (see Theorem 2.2.25, *Implications of Independence, Part III*),

$$\mathrm{E}[Y_i] = \mathrm{E}[Y_i | R_i = 1],$$

and by stable outcomes,

$$\mathrm{E}[Y_i | R_i = 1] = \mathrm{E}[Y_i^* | R_i = 1].$$

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Thus,

$$\mathrm{E}[Y_i] = \mathrm{E}[Y_i^* | R_i = 1]. \ \square$$

Assuming MCAR thus makes it extremely easy to estimate the population mean when we have missing data. The plug-in estimator is just the sample mean of the non-missing values:

$$\hat{\mathrm{E}}[Y_i] = \hat{\mathrm{E}}[Y_i^* | R_i = 1] = \frac{\sum_{i=1}^{n} Y_i^* R_i}{\sum_{i=1}^{n} R_i}.$$

For instance, in Example 6.1.2 (*Missing Data with Binary Outcomes*), our estimate of $\mathrm{E}[Y_i]$ assuming MCAR would be $\frac{3}{4}$. Our ability to draw inferences with missing data thus entirely depends on the strength of the assumptions that we are willing to impose.

One way to think about MCAR is as follows: given a large enough n, we can impute the outcomes for non-responders using the sample mean for responders.

Example 6.1.7. *Imputation under MCAR*
Again, consider Example 6.1.2. We have

Unit	Y_i	R_i
1	1	1
2	?	0
3	1	1
4	0	1
5	1	1
6	?	0

Assuming MCAR, $\mathrm{E}[Y_i | R_i = 0] = \mathrm{E}[Y_i]$, which we can estimate using the above plug-in estimator. Thus, we can impute the missing values as follows:

Unit	Y_i	R_i
1	1	1
2	$\hat{\mathrm{E}}[Y_i] = \frac{3}{4}$	0
3	1	1
4	0	1
5	1	1
6	$\hat{\mathrm{E}}[Y_i] = \frac{3}{4}$	0

Then the sample mean with the missing values imputed is $\frac{3}{4}$.[5] $\triangle$

[5] Recall that, with only six observations, the data are too micronumerous to guarantee that our plug-in estimate would be anywhere near the true value of $\mathrm{E}[Y_i]$. This example is merely illustrative.

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Under the MCAR assumption, imputation is essentially a pointless extra step, since we are using $\hat{\mathrm{E}}[Y_i]$ to impute the missing values in order to get $\hat{\mathrm{E}}[Y_i]$ again. But the utility of this way of thinking about estimating expectations with missing data will become clear in Section 6.2 (*Estimation with Missing Data*).

With the exception of missingness resulting from randomized sampling, the MCAR assumption is unlikely to hold in practice. In the following section, we consider an assumption that incorporates the use of covariates, and is perhaps more palatable in many settings.

6.1.3 Missing at Random [7.1.5]

If we have additional information on each unit, we can impose an alternative, and in many ways weaker, version of the MCAR assumption. This assumption is known as *missing at random (MAR)* or *ignorability*. (We will use the term "MAR" in this chapter; the term "ignorability" will be used for its analog, "strong ignorability," in Section 7.1.5.) MAR is just like MCAR except *conditional on covariates*. (So it is not missing *completely* at random, just missing at random once we condition on covariates.)

Formally, suppose that for each unit i, we observe a vector of covariates $\mathbf{X}_i$. Importantly, we observe $\mathbf{X}_i$ even when $R_i = 0$. That is, we know the covariate values for all units; we only have missing data on the outcome variable Y_i. Now, the random vector of observables is $(Y_i^*, R_i, \mathbf{X}_i)$. Then the MAR assumption looks just like MCAR, except that everything is conditional on $\mathbf{X}_i$.

Definition 6.1.8. *Missing at Random (MAR)*
Let Y_i and R_i be random variables with $\mathrm{Supp}[R_i] = \{0,1\}$, let $Y_i^* = Y_iR_i + (-99)(1-R_i)$, and let $\mathbf{X}_i$ be a random vector. Then Y_i is *missing at random* conditional on $\mathbf{X}_i$ if the following conditions hold:

- $Y_i \perp\!\!\!\perp R_i | \mathbf{X}_i$ (independence of outcome and response conditional on $\mathbf{X}_i$).[6]
- $\exists \epsilon > 0$ such that $\Pr[R_i = 1|\mathbf{X}_i] > \epsilon$ (nonzero probability of response conditional on $\mathbf{X}_i$).[7]

[6] Formally, *conditional independence* is defined as follows: for random variables X, Y, and Z with joint probability mass function/probability density function (PMF/PDF) $f(x,y,z)$, $Y \perp\!\!\!\perp Z|X$ if, $\forall y,z \in \mathbb{R}$ and $\forall x \in \mathrm{Supp}[X]$,

$$f_{(Y,Z)|X}\big((y,z)\,\big|\,x\big) = f_{Y|X}(y|x) f_{Z|X}(z|x).$$

If $Y \perp\!\!\!\perp Z|X$, all the implications of independence derived in Chapters 1 and 2 hold conditional on X; for example, $\mathrm{E}[Y|X] = \mathrm{E}[Y|Z,X]$.

[7] The notation $\Pr[\cdot|X]$ is defined analogously to $\mathrm{E}[\cdot|X]$ and $\mathrm{V}[\cdot|X]$. For example, if we define the function $P_{\{Y=y\}}(x) = \Pr[Y = y|X = x]$, then $\Pr[Y = y|X]$ denotes the random variable

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The first of these conditions is known as a *conditional independence assumption*. Stated in words, the conditional independence assumption says that, among units with the same measured background characteristics, the types that respond and the types that do not respond are exactly the same in terms of their distribution of Y_i values. This assumption is not to be taken lightly, and cannot generally be expected to hold in practice.

Almost every common method of dealing with missing data in applied practice (multiple imputation, reweighting, etc.) depends on some variant of the MAR assumption. The MAR assumption is sufficiently strong to allow us to point identify $\mathrm{E}[Y_i]$.

Theorem 6.1.9. *Expected Value under MAR*
Let Y_i and R_i be random variables with $\mathrm{Supp}[R_i] = \{0,1\}$, let $Y_i^* = Y_i R_i + (-99)(1-R_i)$, and let $\mathbf{X}_i$ be a discrete random vector.[8] Then, if Y_i is MAR conditional on $\mathbf{X}_i$,

$$\mathrm{E}[Y_i] = \sum_{\mathbf{x}} \mathrm{E}[Y_i^* \mid R_i = 1, \mathbf{X}_i = \mathbf{x}] \Pr[\mathbf{X}_i = \mathbf{x}].$$

Proof: By the Law of Iterated Expectations,

$$\mathrm{E}[Y_i] = \mathrm{E}\big[\mathrm{E}[Y_i \mid \mathbf{X}_i]\big] = \sum_{\mathbf{x}} \mathrm{E}[Y_i \mid \mathbf{X}_i = \mathbf{x}] \Pr[\mathbf{X}_i = \mathbf{x}].$$

By MAR (and stable outcomes),

$$\mathrm{E}[Y_i \mid \mathbf{X}_i] = \mathrm{E}[Y_i \mid R_i = 1, \mathbf{X}_i] = \mathrm{E}[Y_i^* \mid R_i = 1, \mathbf{X}_i].$$

Thus,

$$\mathrm{E}[Y_i] = \sum_{\mathbf{x}} \mathrm{E}[Y_i^* \mid R_i = 1, \mathbf{X}_i = \mathbf{x}] \Pr[\mathbf{X}_i = \mathbf{x}]. \quad \square$$

Since we can observe the random vector $(Y_i^*, R_i, \mathbf{X}_i)$, we can point identify $\mathrm{E}[Y_i^* \mid R_i = 1, \mathbf{X}_i = \mathbf{x}]$ and $\Pr[\mathbf{X}_i = \mathbf{x}]$. Thus, under MAR, $\mathrm{E}[Y_i]$ is point identified. Furthermore, note what the second line of the above proof implies that the conditional expectation function (CEF) of Y_i given $\mathbf{X}_i$ is point identified under MAR.

$P_{\{Y=y\}}(X)$. Hence, the statement that $\Pr[R_i = 1 \mid \mathbf{X}_i] > \epsilon$ is equivalent to $\forall \mathbf{x} \in \mathrm{Supp}[\mathbf{X}_i], \Pr[R_i = 1 \mid \mathbf{X}_i = \mathbf{x}] > \epsilon$.

[8] We assume here that $\mathbf{X}_i$ is discrete for ease of exposition. This assumption is not necessary, but it lets us work with PMFs instead of PDFs. The continuous case is analogous. Recall that we are writing $\sum_{\mathbf{x}}$ as a shorthand for $\sum_{\mathbf{x} \in \mathrm{Supp}[\mathbf{X}_i]}$.

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Theorem 6.1.10. *CEF under MAR*
Let Y_i and R_i be random variables with $\text{Supp}[R_i] = \{0,1\}$, let $Y_i^* = Y_iR_i + (-99)(1 - R_i)$, and let $\mathbf{X}_i$ be a random vector. Then, if Y_i is MAR conditional on X_i,

$$\text{E}[Y_i|\mathbf{X}_i] = \text{E}[Y_i^*|R_i = 1, \mathbf{X}_i].$$

A proof directly follows from the second line of the proof of Theorem 6.1.9. Thus, under MAR, the CEF of observable outcomes is identical to the CEF of all outcomes. This result provides some optimism about our ability to approximate and estimate the CEF using only observable quantities. As we will see shortly, the tools developed in Chapter 4 are readily applicable here.

6.1.4 The Role of the Propensity Score for Missing Data [7.1.6]

Before we discuss some other methods of estimating the expected value with missing data, we must first establish a fundamental result. If strong ignorability holds, we need not condition on all the covariates; instead, we can condition on a summary measure of response and the covariates. This summary measure is the *propensity score*, which corresponds to the value of the *response propensity function* evaluated at a unit's covariate values.

Definition 6.1.11. *Response Propensity Function*
Suppose we observe $(R_i, \mathbf{X}_i)$, where $\text{Supp}[R_i] = \{0,1\}$. Then the *response propensity function* is

$$p_R(\mathbf{x}) = \Pr[R_i = 1|\mathbf{X}_i = \mathbf{x}], \forall \mathbf{x} \in \text{Supp}[\mathbf{X}_i].$$

The random variable $p_R(\mathbf{X}_i)$ is known as the propensity score for unit i.[9] The propensity score is the conditional probability of response given the covariates; equivalently, it is the CEF of R_i with respect to $\mathbf{X}_i$ evaluated at $\mathbf{x}$. The propensity score for any given covariate profile is a function of the covariates: if we know that $\mathbf{X}_i = \mathbf{x}$ and we know the full joint distribution of $(R_i, \mathbf{X}_i)$, then we know $\Pr[R_i = 1|\mathbf{X}_i = \mathbf{x}]$.

The following theorem states the key properties of the propensity score under MAR.

Theorem 6.1.12. *MAR and the Propensity Score*
Let Y_i and R_i be random variables with $\text{Supp}[R_i] = \{0,1\}$, let $Y_i^* =$

[9] The term "propensity score" is more commonly used in the context of causal inference. We will discuss this in Chapter 7.

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$Y_iR_i + (-99)(1 - R_i)$, and let $\mathbf{X}_i$ be a random vector. Then, if Y_i is MAR conditional on $\mathbf{X}_i$,

- $R_i \perp\!\!\!\perp \mathbf{X}_i | p_R(\mathbf{X}_i)$. (Balance conditional on $p_R(\mathbf{X}_i)$)
- $Y_i \perp\!\!\!\perp R_i | p_R(\mathbf{X}_i)$. (Independence of outcome and response conditional on $p_R(\mathbf{X}_i)$)
- $\exists \epsilon > 0$ such that $\Pr[R_i = 1 | p_R(\mathbf{X}_i)] > \epsilon$. (Nonzero probability of response conditional on $p_R(\mathbf{X}_i)$)

Proof: We will prove the first two properties and omit a proof of the third.

Balance: Since R_i is binary, its distribution is fully governed by its expected value (Example 5.1.2), so

$$R_i \perp\!\!\!\perp \mathbf{X}_i \big| p_R(\mathbf{X}_i) \iff \mathrm{E}\big[R_i \big| \mathbf{X}_i, p_R(\mathbf{X}_i)\big] = \mathrm{E}\big[R_i \big| p_R(\mathbf{X}_i)\big].$$

Since conditioning on a function of $\mathbf{X}_i$ provides no additional information when we have already conditioned on $\mathbf{X}_i$,

$$\mathrm{E}\big[R_i \big| \mathbf{X}_i, p_R(\mathbf{X}_i)\big] = \mathrm{E}[R_i | \mathbf{X}_i] = p_R(\mathbf{X}_i),$$

and by the Law of Iterated Expectations,

$$\mathrm{E}\big[R_i \big| p_R(\mathbf{X}_i)\big] = \mathrm{E}\big[\mathrm{E}[R_i | \mathbf{X}_i] \big| p_R(\mathbf{X}_i)\big] = \mathrm{E}\big[p_R(\mathbf{X}_i) \big| p_R(\mathbf{X}_i)\big] = p_R(\mathbf{X}_i).$$

Therefore,

$$\mathrm{E}\big[R_i \big| \mathbf{X}_i, p_R(\mathbf{X}_i)\big] = p_R(\mathbf{X}_i) = \mathrm{E}\big[R_i \big| p_R(\mathbf{X}_i)\big],$$

and thus $R_i \perp\!\!\!\perp \mathbf{X}_i | p_R(\mathbf{X}_i)$.

Independence of outcome and response: Since R_i is binary,

$$R_i \perp\!\!\!\perp Y_i \big| p_R(\mathbf{X}_i) \iff \mathrm{E}\big[R_i \big| Y_i, p_R(\mathbf{X}_i)\big] = \mathrm{E}\big[R_i \big| p_R(\mathbf{X}_i)\big].$$

By the Law of Iterated Expectations,

$$\mathrm{E}\big[R_i \big| Y_i, p_R(\mathbf{X}_i)\big] = \mathrm{E}\big[\mathrm{E}[R_i | Y_i, \mathbf{X}_i] \big| Y_i, p_R(\mathbf{X}_i)\big],$$

and by MAR,

$$\mathrm{E}[R_i | Y_i, \mathbf{X}_i] = \mathrm{E}[R_i | \mathbf{X}_i] = p_R(\mathbf{X}_i).$$

So, by substitution,

$$\mathrm{E}\big[R_i \big| Y_i, p_R(\mathbf{X}_i)\big] = \mathrm{E}\big[p_R(\mathbf{X}_i) \big| Y_i, p_R(\mathbf{X}_i)\big] = p_R(\mathbf{X}_i) = \mathrm{E}\big[R_i \big| p_R(\mathbf{X}_i)\big],$$

and thus $R_i \perp\!\!\!\perp Y_i | p_R(\mathbf{X}_i)$. $\square$

In short, Theorem 6.1.12 says that conditioning on the propensity score is equivalent to conditioning on all the covariates. This implies that if Y_i is MAR conditional on $\mathbf{X}_i$, then Y_i is also MAR conditional on just the propensity score $p_R(\mathbf{X}_i)$.

We have shown that, in order to identify the population mean, we do not need to condition on all the covariates as long as we condition on the propensity score. This result is thought to be useful, since estimation is complicated when we attempt to nonparametrically condition on a large number of variables (unless we use a simple approximation like OLS). But there is a problem here: with real data, propensity scores must also be estimated using the same number of variables. If the form of the response propensity function is unknown, then this poses a similar nonparametric problem. We consider this in Section 6.2.4, as the some of the plug-in estimators developed in the next section rely on having estimates that well approximate the true propensity score.

6.2 ESTIMATION WITH MISSING DATA UNDER MAR [7.2]

We have shown that, under the MCAR assumption, a plug-in estimator of $\mathrm{E}[Y_i]$ is simply the sample mean of the observed outcomes. Under MAR, the intuition is much the same, but estimation becomes more complicated. In this section, we discuss several estimators that implement the MAR assumption.

6.2.1 Plug-In Estimation [7.2.1]

If we have discrete covariates, then under the MAR assumption, we can estimate $\mathrm{E}[Y_i]$ with a simple plug-in estimator. Recall from Theorem 6.1.9 that, if $\mathbf{X}_i$ is discrete,

$$\mathrm{E}[Y_i] = \sum_{\mathbf{x}} \mathrm{E}[Y_i^* | R_i = 1, \mathbf{X}_i = \mathbf{x}] \Pr[\mathbf{X}_i = \mathbf{x}].$$

So the plug-in estimator is

$$\begin{aligned} \hat{\mathrm{E}}[Y_i] &= \sum_{\mathbf{x} \in \widehat{\mathrm{Supp}}[\mathbf{X}_i]} \hat{\mathrm{E}}[Y_i^* | R_i = 1, \mathbf{X}_i = \mathbf{x}] \widehat{\Pr}[\mathbf{X}_i = \mathbf{x}] \\ &= \sum_{\mathbf{x} \in \widehat{\mathrm{Supp}}[\mathbf{X}_i]} \frac{\sum_{i=1}^n Y_i^* R_i \cdot \mathrm{I}(\mathbf{X}_i = \mathbf{x})}{\sum_{i=1}^n R_i \cdot \mathrm{I}(\mathbf{X}_i = \mathbf{x})} \cdot \frac{\sum_{i=1}^n \mathrm{I}(\mathbf{X}_i = \mathbf{x})}{n}, \end{aligned}$$

where $\mathrm{I}(\cdot)$ is the indicator function and $\widehat{\mathrm{Supp}}[\mathbf{X}_i]$ is the set of all observed values of $\mathbf{X}_i$ (that is, the plug-in estimator for the support of $\mathbf{X}_i$).[10]

This estimator is sometimes known as the *post-stratification* estimator for missing data—the first ratio is the average observed value of Y_i within a given

10 This requires that we have at least one unit who responded with $\mathbf{X}_i = \mathbf{x}$ for every covariate profile $\mathbf{x} \in \widehat{\mathrm{Supp}}[\mathbf{X}_i]$, otherwise we will have a zero in the denominator. See Footnote 15.

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stratum (the subset of observations in which $\mathbf{X}_i = \mathbf{x}$), and the second ratio is the stratum's share of the overall sample. This is essentially the sample analog to the Law of Iterated Expectations.

Example 6.2.1. *Plug-In Estimation of the Expected Value*
Consider again Example 6.1.2 (*Missing Data with Binary Outcomes*), but now assume that we also observe a binary covariate X_i for every unit.

Unit	Y_i	R_i	X_i
1	1	1	0
2	?	0	0
3	1	1	0
4	0	1	0
5	1	1	1
6	?	0	1

Assuming MAR, the above plug-in estimator yields the estimate

$$\begin{aligned}\hat{\mathrm{E}}[Y_i] &= \hat{\mathrm{E}}[Y_i^* | R_i = 1, X_i = 0]\widehat{\Pr}[X_i = 0] + \hat{\mathrm{E}}[Y_i^* | R_i = 1, X_i = 1]\widehat{\Pr}[X_i = 1] \\ &= \frac{2}{3} \cdot \frac{4}{6} + 1 \cdot \frac{2}{6} = \frac{7}{9}.\end{aligned}$$

Note that this estimate differs from the MCAR-based estimate of $\frac{3}{4}$. We can think of the MAR assumption as allowing us to impute the outcomes for non-responders, only now we impute each missing outcome using the sample mean of respondents with the same covariate values, since under MAR, $\forall \mathbf{x} \in \mathrm{Supp}[\mathbf{X}_i], \mathrm{E}[Y_i | R_i = 0, \mathbf{X}_i = \mathbf{x}] = \mathrm{E}[Y_i | \mathbf{X}_i = \mathbf{x}]$. In Example 6.1.2, this yields

Unit	Y_i	R_i	X_i
1	1	1	0
2	$\hat{\mathrm{E}}[Y_i \vert X_i = 0] = \frac{2}{3}$	0	0
3	1	1	0
4	0	1	0
5	1	1	1
6	$\hat{\mathrm{E}}[Y_i \vert X_i = 1] = 1$	0	1

Then the sample mean with the missing values imputed is $\frac{7}{9}$, which is just the plug-in estimate. △

Similar methods exist for filling in *joint* distributions when we also have missing values for $\mathbf{X}_i$ as well as outcomes. We then need to invoke MAR with respect to all the non-missing variables.

The logic behind this plug-in estimator is also the basis for survey reweighting. Suppose that we have undersampled a certain group characterized by

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$\mathbf{X}_i = \mathbf{x}$, and suppose that we know the population distribution of $\mathbf{X}_i$. If we assume MAR, we can obtain a consistent estimate using the above plug-in estimator.

Example 6.2.2. *Survey Reweighting*
Suppose that we conducted an Internet survey asking a large number of adult U.S. citizens (1) their height (in inches) and (2) their gender. Assume that no one lies or otherwise misreports.[11] Our goal is to estimate the average height of adult Americans. The problem is: people who answer Internet surveys may be unusual, both in ways we can measure and ways we cannot.

Suppose that we obtained results characterized by the following summary statistics:

$$\begin{aligned} &\hat{\mathrm{E}}[Y_i^* | R_i = 1, X_i = 0] = 64, \quad \widehat{\Pr}[X_i = 0 | R_i = 1] = \tfrac{3}{10}, \\ &\hat{\mathrm{E}}[Y_i^* | R_i = 1, X_i = 1] = 70, \quad \widehat{\Pr}[X_i = 1 | R_i = 1] = \tfrac{7}{10}, \end{aligned}$$

where X_i is an indicator variable for gender.[12] The unadjusted estimate of the national average height is

$$\begin{aligned} \hat{\mathrm{E}}[Y_i] &= \hat{\mathrm{E}}[Y_i^* | R_i = 1, X_i = 0]\widehat{\Pr}[X_i = 0 | R_i = 1] \\ &\quad + \hat{\mathrm{E}}[Y_i^* | R_i = 1, X_i = 1]\widehat{\Pr}[X_i = 1 | R_i = 1] \\ &= 64 \cdot \frac{3}{10} + 70 \cdot \frac{7}{10} \\ &= 68.2 \text{ inches.} \end{aligned}$$

This is the estimate we would obtain if we used a plug-in estimator after assuming MCAR, that is, if we used the sample mean as our estimate. But we can do better. Assume that the adult U.S. population is 50% women, so we know that $\Pr[X_i = 0] = \Pr[X_i = 1] = \frac{1}{2}$. Then, under MAR with respect to X_i,

$$\begin{aligned} \mathrm{E}[Y_i] &= \sum_x \mathrm{E}[Y_i | X_i = x] \Pr[X_i = x] \\ &= \sum_x \mathrm{E}[Y_i^* | R_i = 1, X_i = x] \Pr[X_i = x] \\ &= \mathrm{E}[Y_i^* | R_i = 1, X_i = 0] \cdot \frac{1}{2} + \mathrm{E}[Y_i^* | R_i = 1, X_i = 1] \cdot \frac{1}{2}. \end{aligned}$$

Thus, the *adjusted* plug-in estimate of $\mathrm{E}[Y_i]$ is

$$\begin{aligned} \hat{\mathrm{E}}[Y_i] &= \hat{\mathrm{E}}[Y_i^* | R_i = 1, X_i = 0] \cdot \frac{1}{2} + \hat{\mathrm{E}}[Y_i^* | R_i = 1, X_i = 1] \cdot \frac{1}{2} = 64 \cdot \frac{1}{2} + 70 \cdot \frac{1}{2} \\ &= 67 \text{ inches. } \triangle \end{aligned}$$

11 We would recommend that readers not do so outside of this example.
12 By convention, "men are odd": $X_i = 0$ for women and $X_i = 1$ for men. For simplicity in exposition, we assume that non-binary genders were not recorded in this survey.

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It should be clear that the adjusted estimate is more reasonable than the unadjusted estimate. We know that the U.S. population is about 50% women. Additionally, we know (and our data confirm) that gender is associated with height—men are taller on average. So clearly, the distribution of heights in a sample that is 70% men is likely to be unrepresentative of the distribution of heights in the general population. The adjusted estimate allows us to correct for this sampling bias.

However, if MAR conditional on gender does not hold, this adjusted plug-in estimator may still be inconsistent. There might be unobserved (or perhaps even unobservable) determinants of response that are related to height. Is there any reason why Internet survey respondents might tend to be shorter or taller on average than the general population, even conditional on gender?

Thus far, we have dealt with estimation under MAR only in the special case where we have a single discrete covariate that can take on just a small number of values. Once we introduce continuous (or perhaps many) covariates, we cannot use this estimator anymore. The remainder of this chapter considers the many ways in which we might construct estimators when we believe that MAR holds.

6.2.2 Regression Estimation [7.2.2]

Given that we know that the CEF of observable data is equivalent to the CEF of all (not fully observable) data under MAR, this suggests that we might be able to use OLS regression to approximate the CEF of the full data. This approximation can then be used to predict $\mathrm{E}[Y_i]$ using a plug-in estimator. The following example illustrates this idea.

Example 6.2.3. *Addressing Missing Data with OLS Regression*
Once again, consider Example 6.1.2 (*Missing Data with Binary Outcomes*), but now assume that we observe two covariates for every unit:

Unit	Y_i	R_i	$X_{[1]i}$	$X_{[2]i}$
1	1	1	0	3
2	?	0	0	7
3	1	1	0	9
4	0	1	0	5
5	1	1	1	4
6	?	0	1	3

Under MAR, $\forall \mathbf{x} \in \mathrm{Supp}[\mathbf{X}_i], \mathrm{E}[Y_i | R_i = 0, \mathbf{X}_i = \mathbf{x}] = \mathrm{E}[Y_i | \mathbf{X}_i = \mathbf{x}]$, so we want to fill in all the missing values of Y_i with $\hat{\mathrm{E}}[Y_i | X_{[1]i} = x_{[1]}, X_{[2]i} = x_{[2]}]$.

Let us see how we could do this using regression. Suppose we assumed that, at least to a first approximation,

$$\mathrm{E}\big[Y_i \,\big|\, X_{[1]i}, X_{[2]i}\big] = \mathrm{E}\big[Y_i^* \,\big|\, R_i = 1, X_{[1]i}, X_{[2]i}\big] = \beta_0 + \beta_1 X_{[1]i} + \beta_2 X_{[2]i}.$$

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(The first equality is implied by MAR; the second is a functional form assumption.) Then we could estimate the coefficients with ordinary least squares (OLS) and use the resulting equation to impute the missing values:

Unit	Y_i	R_i	$X_{[1]i}$	$X_{[2]i}$
1	1	1	0	3
2	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 7$	0	0	7
3	1	1	0	9
4	0	1	0	5
5	1	1	1	4
6	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot 3$	0	1	3

(Remember: these are just expectations for units with the same covariate values; we cannot *actually* fill in the missing outcomes.) Equivalently, we could instead impute *all* of the Y_i values, which will yield the same estimate for $\mathrm{E}[Y_i]$ and is more straightforward to implement:

Unit	Y_i	R_i	$X_{[1]i}$	$X_{[2]i}$
1	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 3$	1	0	3
2	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 7$	0	0	7
3	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 9$	1	0	9
4	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 5$	1	0	5
5	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot 4$	1	1	4
6	$\hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot 3$	0	1	3

In either case, the estimate of $\mathrm{E}[Y_i]$ is then the sample mean with imputed values for potential outcomes. This estimator would be consistent if MAR held and if the functional form of the CEF were in fact $\mathrm{E}[Y_i^* | R_i = 1, X_{[1]i}, X_{[2]i}] = \beta_0 + \beta_1 X_{[1]i} + \beta_2 X_{[2]i}$. $\triangle$

The above example described the case of estimating a linear approximation to the CEF. We could have considered nonlinear (Section 4.3) specifications, including interactions or polynomials. Once we have an estimate of the CEF, we can define the following plug-in estimator for $\mathrm{E}[Y_i]$.

Definition 6.2.4. *Regression Estimator for Missing Data*
Let Y_i and R_i be random variables with $\mathrm{Supp}[R_i] = \{0, 1\}$. Let $Y_i^* = Y_i R_i + (-99)(1 - R_i)$ and let $\mathbf{X}_i$ be a random vector. Then, given n i.i.d. observations of $(Y_i^*, R_i, \mathbf{X}_i)$, the *regression estimator* for $\mathrm{E}[Y_i]$ is

$$\hat{\mathrm{E}}_R[Y_i] = \frac{1}{n} \sum_{i=1}^{n} \hat{\mathrm{E}}[Y_i^* | R_i = 1, \mathbf{X}_i],$$

where $\hat{\mathrm{E}}[Y_i^* | R_i = 1, \mathbf{X}_i = \mathbf{x}]$ is an estimator of the CEF.

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As the regression estimator is a plug-in estimator, we have a good sense of its theoretical properties: the estimator will behave well when the CEF is well characterized by the specification of the regression and the usual plug-in regularity conditions hold.

6.2.3 Hot Deck Imputation [7.2.4]

Another method of imputing missing outcomes is *hot deck imputation*, using nearest-neighbor matching with the propensity score (which, for the moment, we assume is known). We begin by discussing hot deck imputation because it is one of the simplest approaches, and because it will convey broader intuitions about imputation estimators. The following example illustrates hot deck imputation.

Example 6.2.5. *Imputation with the Propensity Score*
Suppose that we had the following data:

Unit	Y_i	R_i	$X_{[1]i}$	$X_{[2]i}$	$p_R(\mathbf{X}_i)$
1	2	1	0	3	0.33
2	?	0	0	7	0.14
3	3	1	0	9	0.73
4	10	1	0	5	0.35
5	12	1	1	4	0.78
6	?	0	1	3	0.70

Assume that MAR holds and that our goal is to estimate $\mathrm{E}[Y_i]$. By Theorem 6.1.12 (*MAR and the Propensity Score*), the MAR assumption implies that $\mathrm{E}[Y_i|R_i=0, p_R(\mathbf{X}_i)] = \mathrm{E}[Y_i|p_R(\mathbf{X}_i)]$, so we can impute the missing outcomes with estimates of $\mathrm{E}[Y_i|p_R(\mathbf{X}_i)]$:

Unit	Y_i	R_i	$X_{[1]i}$	$X_{[2]i}$	$p_R(\mathbf{X}_i)$
1	2	1	0	3	0.33
2	$\hat{\mathrm{E}}[Y_i\|p_R(\mathbf{X}_i)=0.14]$	0	0	7	0.14
3	3	1	0	9	0.73
4	10	1	0	5	0.35
5	12	1	1	4	0.78
6	$\hat{\mathrm{E}}[Y_i\|p_R(\mathbf{X}_i)=0.70]$	0	1	3	0.70

Then, if our estimator of $\mathrm{E}[Y_i|p_R(\mathbf{X}_i)=p_R(\mathbf{x})]$ is consistent, the sample mean with the missing values imputed will be a consistent estimator of the population mean.

Hot deck imputation is perhaps one of the simplest imputation estimators: for each missing unit, find the observed unit that is closest on $p_R(\mathbf{X}_i)$ and

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use that unit's outcome to fill in the missing outcome.[13] In this example, we have two unobserved outcomes to impute. Unit 2 has the propensity score $p_R(\mathbf{X}_2) = 0.14$. The observed unit with the closest propensity score is unit 1 with $p_R(\mathbf{X}_1) = 0.33$, so we use $Y_1 = 2$ to impute Y_2. Likewise, unit 6 has the propensity score $p_R(\mathbf{X}_6) = 0.70$, and the observed units with closest propensity score is unit 3 with $p_R(\mathbf{X}_3) = 0.73$, so we use $Y_3 = 3$ to impute Y_6. This yields

Unit	Y_i	R_i	$X_{[1]i}$	$X_{[2]i}$	$p_R(\mathbf{X}_i)$
1	2	1	0	3	0.33
2	2	0	0	7	0.14
3	3	1	0	9	0.73
4	10	1	0	5	0.35
5	12	1	1	4	0.78
6	3	0	1	3	0.70

So the hot deck imputation estimate of $\mathrm{E}[Y_i]$ is $\frac{32}{6} = \frac{16}{3} \approx 5.33$. △

We would hesitate to recommend the use of hot deck imputation in practice. Intuitively, hot deck imputation discards a great deal of information about the data. In the above example, we used $Y_1 = 2$ to impute Y_2 because $p_R(\mathbf{X}_1) = 0.33$ is closest to $p_R(\mathbf{X}_2) = 0.14$. But $p_R(\mathbf{X}_4) = 0.35$ is almost as close to $p_R(\mathbf{X}_1)$, so $Y_4 = 10$ probably also gives us some information about $\mathrm{E}[Y_i | p_R(\mathbf{X}_i) = 0.14]$. So why would we ignore Y_4 entirely and just use Y_1 as our guess for Y_2? This seems hard to justify.

Furthermore, in order to use hot deck imputation based on the propensity score, we need to have estimates of the propensity scores to work with. We take up this issue in the following section.

6.2.4 Maximum Likelihood Plug-In Estimation of Propensity Scores [7.2.5]

We can also use ML to estimate unknown propensity scores, which can in turn be used in estimators like hot deck imputation (with the caveats listed above) and the weighting estimators that we will consider shortly. Recall from Definition 6.1.11 that the propensity score $p_R(\mathbf{x}) = \Pr[R_i = 1 | \mathbf{X}_i = \mathbf{x}]$, and, from Example 2.1.3, that $\Pr[R_i = 1 | \mathbf{X}_i = \mathbf{x}] = \mathrm{E}[R_i | \mathbf{X}_i = \mathbf{x}]$. Therefore, estimating the response propensity function is equivalent to estimating the CEF of R_i with respect to $\mathbf{X}_i$. However, this is a case where OLS would not be suitable for estimating an unknown CEF. We know that $\mathrm{Supp}[\mathrm{E}[R_i | \mathbf{X}_i]] \in [0, 1]$, and thus we might be concerned that a linear approximation would yield impossible propensity score estimates (predictions greater than zero or less than one).

13 The name "hot deck imputation" derives from the era when data were stored on punch cards. To impute missing outcomes, one would use an observed value from the same dataset—a stack of cards—which was still hot from being processed.

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We know, however, that if we assume a binary choice model, it will respect the constraints on the distribution of R_i.

Suppose now that we have a binary choice model for response with mean function h. If this model is true, then following Section 5.2.4, we have can use a ML estimator to obtain propensity score estimates: $\hat{p}_R(\mathbf{X}_i) = h(\mathbf{X}_i\hat{\boldsymbol{\beta}}_{ML})$. If the binary choice model is true, then, under the usual plug-in regularity conditions, $\hat{p}_R(\mathbf{x})$ will be consistent for $p_R(\mathbf{x})$. If the model is not true, then our estimates of the propensity score will only be as good as our approximation permits. Of course, if data permit, we could fit a more complex approximation, perhaps altering how $\mathbf{X}_i$ is specified (to include, say, polynomials or interactions) or using a mixture model (Section 5.2.5). Regardless of the exact specification, we will assume as we proceed that we are using an ML plug-in estimate from a binary choice model, $\hat{p}_R(\mathbf{x})$, for our plug-in estimates, though our results do not depend on this.

6.2.5 Weighting Estimators [7.2.6]

We are not limited to hot deck imputation in order to use the propensity score to estimate $\mathrm{E}[Y_i]$. An alternative is provided by *inverse probability–weighted (IPW) estimators*. Before we can describe these estimators, we first establish an important population result. Transforming the observed outcomes by weighting by $\frac{1}{p_R(\mathbf{X}_i)}$ allows us to express the expected value of the (not fully observable) data as an expected value of the (reweighted) observable data.

Theorem 6.2.6. *Reweighting under MAR*
Let Y_i and R_i be random variables with $\mathrm{Supp}[R_i] = \{0,1\}$. Let $Y_i^* = Y_iR_i + (-99)(1-R_i)$ and let $\mathbf{X}_i$ be a random vector. Then, if R_i is MAR conditional on $\mathbf{X}_i$,

$$\mathrm{E}[Y_i] = \mathrm{E}\left[\frac{Y_i^* R_i}{p_R(\mathbf{X}_i)}\right].$$

Proof: By definition,

$$\mathrm{E}\left[\frac{Y_i^* R_i}{p_R(\mathbf{X}_i)}\middle|\mathbf{X}_i\right] = \mathrm{E}\left[Y_i^* \cdot \frac{R_i}{\Pr[R_i = 1|\mathbf{X}_i]}\middle|\mathbf{X}_i\right],$$

and by stable outcomes,

$$\mathrm{E}\left[Y_i^* \cdot \frac{R_i}{\Pr[R_i = 1|\mathbf{X}_i]}\middle|\mathbf{X}_i\right] = \mathrm{E}\left[Y_i \cdot \frac{R_i}{\Pr[R_i = 1|\mathbf{X}_i]}\middle|\mathbf{X}_i\right].$$

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Then, since MAR holds, $Y_i \perp\!\!\!\perp R_i | \mathbf{X}_i$, so

$$\begin{aligned}
\mathrm{E}\left[Y_i \cdot \frac{R_i}{\Pr[R_i = 1 | \mathbf{X}_i]} \middle| \mathbf{X}_i\right] &= \mathrm{E}[Y_i | \mathbf{X}_i] \cdot \mathrm{E}\left[\frac{R_i}{\Pr[R_i = 1 | \mathbf{X}_i]} \middle| \mathbf{X}_i\right] \\
&= \mathrm{E}[Y_i | \mathbf{X}_i] \cdot \frac{\mathrm{E}[R_i | \mathbf{X}_i]}{\Pr[R_i = 1 | \mathbf{X}_i]} \\
&= \mathrm{E}[Y_i | \mathbf{X}_i] \cdot \frac{\Pr[R_i = 1 | \mathbf{X}_i]}{\Pr[R_i = 1 | \mathbf{X}_i]} \\
&= \mathrm{E}[Y_i | \mathbf{X}_i].
\end{aligned}$$

Thus,

$$\mathrm{E}\left[\frac{Y_i^* R_i}{p_R(\mathbf{X}_i)} \middle| \mathbf{X}_i\right] = \mathrm{E}[Y_i | \mathbf{X}_i].$$

So, by the Law of Iterated Expectations,

$$\begin{aligned}
\mathrm{E}[Y_i] &= \mathrm{E}\big[\mathrm{E}[Y_i | \mathbf{X}_i]\big] \\
&= \mathrm{E}\left[\mathrm{E}\left[\frac{Y_i^* R_i}{p_R(\mathbf{X}_i)} \middle| \mathbf{X}_i\right]\right] \\
&= \mathrm{E}\left[\mathrm{E}\left[\frac{Y_i R_i}{p_R(\mathbf{X}_i)} \middle| \mathbf{X}_i\right]\right] \\
&= \mathrm{E}\left[\frac{Y_i R_i}{p_R(\mathbf{X}_i)}\right]. \quad \square
\end{aligned}$$

The IPW estimator is again a plug-in estimator: we replace $p_R(\mathbf{X}_i)$ with an estimate thereof, $\hat{p}_R(\mathbf{X}_i)$ (as in the previous section), and substitute a sample mean for an expected value.

Definition 6.2.7. *Inverse Probability–Weighted (IPW) Estimator for Missing Data*
Let Y_i and R_i be random variables with $\mathrm{Supp}[R_i] = \{0,1\}$. Let $Y_i^* = Y_i R_i + (-99)(1 - R_i)$ and let $\mathbf{X}_i$ be a random vector. Then, given n i.i.d. observations of $(Y_i^*, R_i, \mathbf{X}_i)$, the *inverse probability–weighted estimator* for $\mathrm{E}[Y_i]$ is

$$\hat{\mathrm{E}}_{IPW}[Y_i] = \frac{1}{n} \sum_{i=1}^{n} \frac{Y_i^* R_i}{\hat{p}_R(\mathbf{X}_i)}.$$

The earliest estimator of this kind was derived for the case of survey sampling with known response probabilities (Horvitz and Thompson 1952). The

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logic for IPW estimators is intuitively the same as in Example 6.2.2 (*Survey Reweighting*). Suppose that some units are unlikely to offer a response, and so they have small propensity scores. As a result, these types of units are under-represented in the observable data; accordingly, we would want to give the units that we *do* see greater weight. In the case where there is a discrete and finite number of values for $\mathbf{X}_i$, the IPW estimator is logically equivalent to the plug-in estimator from Section 6.2.1 (*Plug-In Estimation*).

The IPW estimator has high variability (that is, its standard error is large) with small samples. An alternative that performs better in practice is a "ratio estimator" proposed by Hájek (1971), known as the stabilized IPW estimator.

Definition 6.2.8. *Stabilized IPW Estimator for Missing Data*
Let Y_i and R_i be random variables with $\text{Supp}[R_i] = \{0,1\}$. Let $Y_i^* = Y_iR_i + (-99)(1-R_i)$ and let $\mathbf{X}_i$ be a random vector. Then, given n i.i.d. observations of $(Y_i^*, R_i, \mathbf{X}_i)$, the *stabilized IPW estimator* for $\text{E}[Y_i]$ is

$$\hat{\text{E}}_{SIPW}[Y_i] = \frac{\frac{1}{n}\sum_{i=1}^{n}\frac{Y_iR_i}{\hat{p}_R(\mathbf{X}_i)}}{\frac{1}{n}\sum_{i=1}^{n}\frac{R_i}{\hat{p}_R(\mathbf{X}_i)}}.$$

When $\hat{p}_R(\mathbf{X}_i)$ converges (in an appropriate functional sense) to $p_R(\mathbf{X}_i)$, then the denominator of the fraction will converge in probability to 1. To see this, note that with the true propensity score, $\text{E}[\frac{R_i}{p_R(\mathbf{X}_i)}] = 1$. However, this normalization provides benefits in estimation. In the case where we draw an unusually large number of units where $p_R(\mathbf{X}_i)$ is large or small, the stabilized IPW estimator's denominators adjust accordingly. Because the stabilized IPW estimator accounts for this between-sample variation in $\sum_{i=1}^{n}\frac{1}{p_R(\mathbf{X}_i)}$, it typically confers efficiency gains over the IPW estimator in practice. Valid asymptotic inference for IPW estimators can be achieved under strong ignorability, proper specification of $\hat{p}_R(\mathbf{X}_i)$, and the usual plug-in regularity conditions via the bootstrap, re-estimating the propensity scores in each bootstrap sample.

6.2.6 Doubly Robust Estimators [7.2.7]

We are not forced to decide between regression estimators and weighting estimators; instead, we can use both using a *doubly robust (DR)* estimator. Under MAR, the DR estimator yields consistent estimates when *either* the regression specification *or* the propensity score specification is correct. Before we state this result, we note an important population result.

Theorem 6.2.9. *Double Robustness Theorem for Missing Data*
Let Y_i and R_i be random variables with $\text{Supp}[R_i] = \{0,1\}$. Let $Y_i^* = Y_i R_i + (-99)(1 - R_i)$ and let $\mathbf{X}_i$ be a random vector. Let $\tilde{\text{E}}[Y_i | R_i = 1, \mathbf{X}_i = \mathbf{x}]$ be an approximation of the CEF evaluated at $R_i = 1$ and $\mathbf{X}_i = \mathbf{x}$, and let $\tilde{p}_R(\mathbf{X}_i)$ be an approximation of the propensity score. If MAR holds and if either

(i) $\tilde{\text{E}}[Y_i^* | R_i = 1, \mathbf{X}_i = \mathbf{x}] = \text{E}[Y_i^* | R_i = 1, \mathbf{X}_i = \mathbf{x}], \forall \mathbf{x} \in \text{Supp}[\mathbf{X}_i]$, and $\exists \epsilon > 0$ such that $\epsilon < \tilde{p}_R(\mathbf{x}) < 1 - \epsilon$,

or

(ii) $\tilde{p}_R(\mathbf{x}) = p_R(\mathbf{x}), \forall \mathbf{x} \in \text{Supp}[\mathbf{X}_i]$,

then

$$\text{E}[Y_i] = \text{E}\left[\tilde{\text{E}}[Y_i^* | R_i = 1, \mathbf{X}_i] + \frac{R_i \left(Y_i^* - \tilde{\text{E}}[Y_i^* | R_i = 1, \mathbf{X}_i] \right)}{\tilde{p}_R(\mathbf{X}_i)} \right].$$

Proof: To consider case (i), suppose that $\tilde{\text{E}}[Y_i^* | R_i = 1, \mathbf{X}_i = \mathbf{x}] = \text{E}[Y_i^* | R_i = 1, \mathbf{X}_i = \mathbf{x}], \forall \mathbf{x} \in \text{Supp}[\mathbf{X}_i]$. Then we have

$$\begin{aligned}
&\text{E}\left[\tilde{\text{E}}[Y_i^* | R_i = 1, \mathbf{X}_i] + \frac{R_i \left(Y_i^* - \tilde{\text{E}}[Y_i^* | R_i = 1, \mathbf{X}_i] \right)}{\tilde{p}_R(\mathbf{X}_i)} \right] \\
&\quad = \text{E}\left[\text{E}[Y_i^* | R_i = 1, \mathbf{X}_i] + \frac{R_i \left(Y_i^* - \text{E}[Y_i^* | R_i = 1, \mathbf{X}_i] \right)}{\tilde{p}_R(\mathbf{X}_i)} \right] \\
&\quad = \text{E}\big[\text{E}[Y_i^* | R_i = 1, \mathbf{X}_i] \big] + \text{E}\left[\frac{R_i Y_i^*}{\tilde{p}_R(\mathbf{X}_i)} \right] - \text{E}\left[\frac{R_i \text{E}[Y_i^* | R_i = 1, \mathbf{X}_i]}{\tilde{p}_R(\mathbf{X}_i)} \right] \\
&\quad = \text{E}\big[\text{E}[Y_i^* | R_i = 1, \mathbf{X}_i] \big] + \text{E}\left[\frac{R_i \text{E}[Y_i^* | R_i = 1, \mathbf{X}_i]}{\tilde{p}_R(\mathbf{X}_i)} \right] \\
&\qquad - \text{E}\left[\frac{R_i \text{E}[Y_i^* | R_i = 1, \mathbf{X}_i]}{\tilde{p}_R(\mathbf{X}_i)} \right] \\
&\quad = \text{E}\big[\text{E}[Y_i^* | R_i = 1, \mathbf{X}_i] \big] \\
&\quad = \text{E}[Y_i],
\end{aligned}$$

where the third equality follows from the Law of Iterated Expectations and the fifth follows from MAR (see Theorem 6.1.9).

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To consider case (ii), suppose instead that $\tilde{p}_R(\mathbf{x}) = p_R(\mathbf{x}), \forall \mathbf{x} \in \text{Supp}[\mathbf{X}_i]$. Then

$$\begin{aligned}
&\mathrm{E}\left[\tilde{\mathrm{E}}[Y_i^*|R_i=1,\mathbf{X}_i] + \frac{R_i\left(Y_i^* - \tilde{\mathrm{E}}[Y_i^*|R_i=1,\mathbf{X}_i]\right)}{\tilde{p}_R(\mathbf{X}_i)}\right] \\
&\quad = \mathrm{E}\left[\tilde{\mathrm{E}}[Y_i^*|R_i=1,\mathbf{X}_i] + \frac{R_i\left(Y_i^* - \tilde{\mathrm{E}}[Y_i^*|R_i=1,\mathbf{X}_i]\right)}{p_R(\mathbf{X}_i)}\right] \\
&\quad = \mathrm{E}\Big[\tilde{\mathrm{E}}[Y_i^*|R_i=1,\mathbf{X}_i]\Big] + \mathrm{E}\left[\frac{R_iY_i^*}{p_R(\mathbf{X}_i)}\right] - \mathrm{E}\left[\frac{R_i\tilde{\mathrm{E}}[Y_i^*|R_i=1,\mathbf{X}_i]}{p_R(\mathbf{X}_i)}\right] \\
&\quad = \mathrm{E}\Big[\tilde{\mathrm{E}}[Y_i^*|R_i=1,\mathbf{X}_i]\Big] + \mathrm{E}\left[\frac{R_iY_i^*}{p_R(\mathbf{X}_i)}\right] - \mathrm{E}\Big[\tilde{\mathrm{E}}[Y_i^*|R_i=1,\mathbf{X}_i]\Big] \\
&\quad = \mathrm{E}\left[\frac{R_iY_i^*}{p_R(\mathbf{X}_i)}\right] \\
&\quad = \mathrm{E}[Y_i]. \ \square
\end{aligned}$$

The Double Robustness Theorem establishes that if either the approximation of the CEF or the approximation of the response propensity function are exactly correct, then we can write the expected value of the full (only partially observable) data in terms of simple expected values of the observable data and the approximations. This result motivates a plug-in estimator, obtained by replacing expected values with sample means, $\tilde{\mathrm{E}}[Y_i|R_i=1,\mathbf{X}_i=\mathbf{x}]$ with some estimate $\hat{\mathrm{E}}[Y_i|R_i=1,\mathbf{X}_i=\mathbf{x}]$, and $\tilde{p}_R(\mathbf{X}_i)$ with some estimate $\hat{p}_R(\mathbf{X}_i)$.

Definition 6.2.10. *Doubly Robust (DR) Estimator for Missing Data*
Let Y_i and R_i be random variables with $\text{Supp}[R_i] = \{0,1\}$. Let $Y_i^* = Y_iR_i + (-99)(1-R_i)$ and let $\mathbf{X}_i$ be a random vector. Then, given n i.i.d. observations of $(Y_i^*, R_i, \mathbf{X}_i)$, the *doubly robust estimator* is

$$\hat{\mathrm{E}}_{DR}[Y_i] = \frac{1}{n}\sum_{i=1}^{n}\hat{\mathrm{E}}[Y_i^*|R_i=1,\mathbf{X}_i] + \frac{1}{n}\sum_{i=1}^{n}\frac{R_i\left(Y_i^* - \hat{\mathrm{E}}[Y_i^*|R_i=1,\mathbf{X}_i]\right)}{\hat{p}_R(\mathbf{X}_i)},$$

where $\hat{\mathrm{E}}[Y_i^*|R_i=1,\mathbf{X}_i]$ is an estimator for $\mathrm{E}[Y_i^*|R_i=1,\mathbf{X}_i]$, and $\hat{p}_R(\mathbf{X}_i)$ is an estimator for $p_R(\mathbf{X}_i)$.

The DR estimator can be decomposed into two elements. The first term, $\frac{1}{n}\sum_{i=1}^{n}\hat{\mathrm{E}}[Y_i|R_i=1,\mathbf{X}_i]$, is a standard regression estimator from Section 6.2.2

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(*Regression Estimation*). The second term, $\frac{1}{n}\sum_{i=1}^{n}\frac{R_i(Y_i-\hat{\mathrm{E}}[Y_i|R_i=1,\mathbf{X}_i])}{\hat{p}_R(\mathbf{X}_i)}$, is an IPW estimator correcting for any unusual deviations in the actual data from the regression estimate. As a plug-in estimator, if the conditions for Theorem 6.2.9 and the usual plug-in regularity conditions hold, the DR estimator can be used in the usual manner, with the bootstrap available for inference. Insofar as the MAR assumption can be believed, the DR estimator provides a sensible and flexible approach that combines the virtues of regression and IPW approaches to estimation.

6.3 APPLICATION: ESTIMATING THE CROSS-NATIONAL AVERAGE OF CLEAN ENERGY USE

We now consider an application, again using the Quality of Governance data (Teorell et al. 2016). Suppose we are interested in the average rate of use of alternative and nuclear energy at the country level. Alternative and nuclear energy use is measured as the percentage of total energy production in a country generated from non–carbon dioxide emitting sources, including, but not limited to, nuclear, wind, hydroelectric, geothermal, and solar power. However, of the 184 countries in the dataset, 50 countries are missing data on this variable. Our goal in this application is to compute estimates of the average using the identifying assumptions and methods detailed in this chapter.

We assume that our observed data are i.i.d. draws of (Y_i^*, R_i), where Y_i^* is equal to country i's percentage use of alternative and nuclear energy if observed and -99 if not observed, and R_i is an indicator for whether country i's rate of use of alternative and nuclear energy is observed. Our goal is to estimate $\mathrm{E}[Y_i]$. We begin with the most credible approach: estimating sharp bounds, as discussed in Section 6.1.1 (*Bounds*). Since percentages are naturally bounded, the lowest and highest possible values for Y_i are 0 and 100, respectively. To estimate lower and upper bounds, we use the usual plug-in estimator, and compute standard errors for these quantities using the bootstrap. Without making further assumptions, our estimated bounds are consistent for the narrowest possible bounds on $\mathrm{E}[Y_i]$ given the observable data (Y_i^*, R_i). These bounds are quite wide, leaving us with a large range of possibilities: our estimates imply that the true value of $\mathrm{E}[Y_i]$ is likely to be somewhere between 7.110 and 34.283. Nevertheless, these estimates are not completely uninformative: accounting for sampling variability, a 95% confidence interval computed using our lower bound estimate excludes all values below 5.109, suggesting that we can be quite confident that the cross-national average rate of clean energy use is above 5%.

We can achieve point identification of $\mathrm{E}[Y_i]$ under stronger assumptions, at the cost of reduced credibility. The simplest such assumption is MCAR—assuming that the population distribution of outcomes for countries for which we have data is the same as the population distribution of outcomes for

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countries for which we do not have data: $Y_i \perp\!\!\!\perp R_i$. This additional assumption gives us point identification, and yields a natural plug-in estimator: we take the sample mean for countries with observed outcomes, so that the MCAR plug-in estimator, $\hat{\mathrm{E}}[Y_i] = \hat{\mathrm{E}}[Y_i^* | R_i = 1] = 9.762$, and an associated 95% confidence interval is $(7.157, 12.367)$. However, the MCAR assumption may not be credible in this setting. The types of countries with observed values for clean energy use may differ systematically from the types of countries with missing values. For example, we might think that more developed countries would have both higher rates of clean energy use as well as better reporting standards. For this reason, we now turn to using covariates to estimate $\mathrm{E}[Y_i]$ under more reasonable assumptions.

We now assume that the data are i.i.d. draws of $(Y_i^*, R_i, \mathbf{X}_i)$, where $\mathbf{X}_i$ is a vector of covariates. In this setting, we have in $\mathbf{X}_i$ the infant mortality rate, urban population, population with electricity, an indicator denoting whether the country is classified as a democracy, population growth rate, GDP/capita, Internet use, pollution levels (as measured by particulate matter), educational attainment for women ages 15–24, and access to clean water.[14] We could now assume MAR with respect to $\mathbf{X}_i$. This implies that, conditional on the values of the observed explanatory variables in $\mathbf{X}$, the distributions of outcomes for countries for which we have response data are exactly the same as the distribution of outcomes for countries for which we do not have response data: $Y_i \perp\!\!\!\perp R_i | \mathbf{X}_i$. While we might think that the estimates obtained by explicitly accounting for the observed explanatory variables are a better approximation of the truth, there is no such guarantee. For example, suppose that countries will only report their clean energy usage if it crosses some country-specific threshold unobservable to us. Then our estimates of $\mathrm{E}[Y_i]$ assuming MAR will typically overestimate energy usage, and no conditioning strategy that fails to account for this (unobservable) threshold can be guaranteed to recover the truth.

That being said, if we assume MAR, we have a variety of estimators available to us. We first use two variants of regression estimation to

[14] Infant mortality rate is measured as the number of infants who died before reaching one year of age per 1,000 live births in a given year; urban population is measured as the percentage of the population living in urban areas as defined by national statistical offices; population with electricity is measured as the percentage of the population with access to electricity; the democracy variable is coded as one if the country is a democracy and zero otherwise; population growth rate is measured as the annual population growth in percentage points; GDP/capita is measured as gross domestic product in current U.S. dollars divided by midyear population; Internet use is measured as the number of individuals who have used the Internet in the last 12 months; particulate matter is measured as population-weighted three-year average exposure to fine particulate matter ($PM_{2.5}$); educational attainment for young women is measured as the average years of education for women ages 15–24; and access to clean water is measured as the percentage of the population with access to a source for clean drinking water. For further details, please refer to the Quality of Governance Standard Data codebook (Teorell et al. 2016).

TABLE 6.3.1. *Estimates of the Cross-National Average of Clean Energy Usage with Missing Data*

	Estimate	Bootstrap SE
Lower Bound Plug-In	7.110	1.021
Upper Bound Plug-In	34.283	3.128
MCAR Plug-In	9.762	1.329
OLS (Linear)	8.997	1.203
OLS (Polynomial)	8.084	1.260
Hot Deck Imputation	8.538	1.204
IPW	8.361	1.111
Stabilized IPW	8.737	1.154
Doubly Robust (Linear)	8.575	1.151
Doubly Robust (Polynomial)	8.716	1.187

impute missing responses. As we might expect, the first regression uses OLS under the working assumption that the CEF of the observed data is linear, $\mathrm{E}[Y_i^*|R_i, \mathbf{X}_i] \approx \beta_0 + \mathbf{X}_i\boldsymbol{\beta}$. The resulting estimate, denoted OLS (linear), is $\hat{\mathrm{E}}[Y_i] = 8.997$, somewhat lower than the MCAR plug-in estimate. We also use a more flexible specification, with $\mathrm{E}[Y_i^*|R_i, \mathbf{X}_i] \approx \beta_0 + \mathbf{X}_i\boldsymbol{\beta} + \mathbf{X}_{[2]i}\boldsymbol{\beta}_2$, where $\mathbf{X}_{[2]i}$ is a vector consisting of the squared values of each element $\mathbf{X}_i$. Estimating $\mathrm{E}[Y_i^*|R_i, \mathbf{X}_i]$ with OLS, we compute the OLS (polynomial) estimate $\hat{\mathrm{E}}[Y_i] = 8.084$, again, lower than the MCAR-based estimate.

In order to use the propensity score-based estimators, we need to first estimate the propensity scores, $p_R(\mathbf{X}_i) = \Pr[R_i = 1|\mathbf{X}_i]$. Doing so is not as simple as using the regression-based approaches, as we would like to ensure that $p_R(\mathbf{X}_i)$ obeys the natural constraints given on probabilities, namely that it lies in the unit interval. To do so, we impose as a working assumption that $p_R(\mathbf{X}_i) \approx \frac{e^{\mathbf{X}_i\Gamma}}{(1+e^{\mathbf{X}_i\Gamma})}$, or a *logit* binary choice model (see Section 5.1.2, *Binary Choice Model*, for more details). We estimate each $\hat{p}_R(\mathbf{X}_i)$ using maximum likelihood (ML) estimation (see Section 5.2, *Maximum Likelihood Estimation*). Our estimate using one-to-one, nearest-neighbor hot deck imputation based on the propensity score largely agrees with that of the regression estimates, with $\hat{\mathrm{E}}[Y_i] = 8.538$.[15]

[15] The bootstrap may not work as a method for computing valid confidence intervals for hot deck imputation since the bias of the estimator may converge slowly (see, e.g., Abadie and Imbens 2008), but the bootstrap standard errors are likely to adequately characterize sampling variability.

We now turn to the IPW and DR estimators.[16] In this setting, we estimate that the IPW and DR estimators are the most precise of the MAR-based estimates, with bootstrap standard errors lower than those of the pure imputation-based approaches. The IPW estimator yields an estimate of $\hat{\mathrm{E}}[Y_i] = 8.361$, while the stabilized IPW estimator yields an estimate of $\hat{\mathrm{E}}[Y_i] = 8.737$. We implement two variants of the DR estimator: one using the OLS (linear) specification for imputation, yielding an estimate $\hat{\mathrm{E}}[Y_i] = 8.575$, and one using the OLS (polynomial) specification for imputation, yielding an estimate $\hat{\mathrm{E}}[Y_i] = 8.716$.

Table 6.3.1 summarizes these estimates. All our MAR-based estimates roughly agree, lying between 8 and 9 percent, with similar levels of uncertainty across estimates. Regardless of the specification, our MAR-based estimates (i) as expected, lie within the estimated bounds and (ii) are lower than the MCAR-based estimator. If we were to believe the MAR assumption, this would imply that countries with missing outcome data are, on average, less likely to use clean energy sources than are countries for which data are available. This finding, however, depends crucially on the assumption of MAR. And MAR remains unverifable, with no clear theoretical heuristic for ascertaining its validity in this setting. The agnostic analyst must interpret these estimates with appropriate skepticism.

6.4 FURTHER READINGS

Tsiatis (2006) provides a now-canonical book-length treatment of missing data, and links it to semiparametric theory elegantly—the book is strongly recommended. For discussions of partial identification of distributional features with missing data (and other partially revealed models), we strongly recommend Manski (2003) and Manski (2013) as comprehensive books; Imbens (2013) provides a valuable critical review of the latter. Rosenbaum (1987) provides an excellent discussion of the properties of IPW-style estimators. Kang and Schafer (2007) provides an informative review piece on the properties and history of doubly robust estimators. Van der Laan and Robins (2003) provide a highly technical book-length treatment of doubly robust estimation, building on earlier work from Robins and coauthors; e.g., Robins, Rotnitzky, and Zhao (1995) and Robins (1999). Bang and Robins (2005) offers an elegant and moderately technical article-length treatment of doubly robust estimation.

[16] As discussed in Section 7.3.2, we Winsorize our propensity score estimates to lie within $[0.01, 1]$.